# Intelli-Credit

AI-Powered Corporate Credit Decisioning Engine

Resolving the 'Data Paradox' for Indian Corporate Lending

# The Problem: The 'Data Paradox'

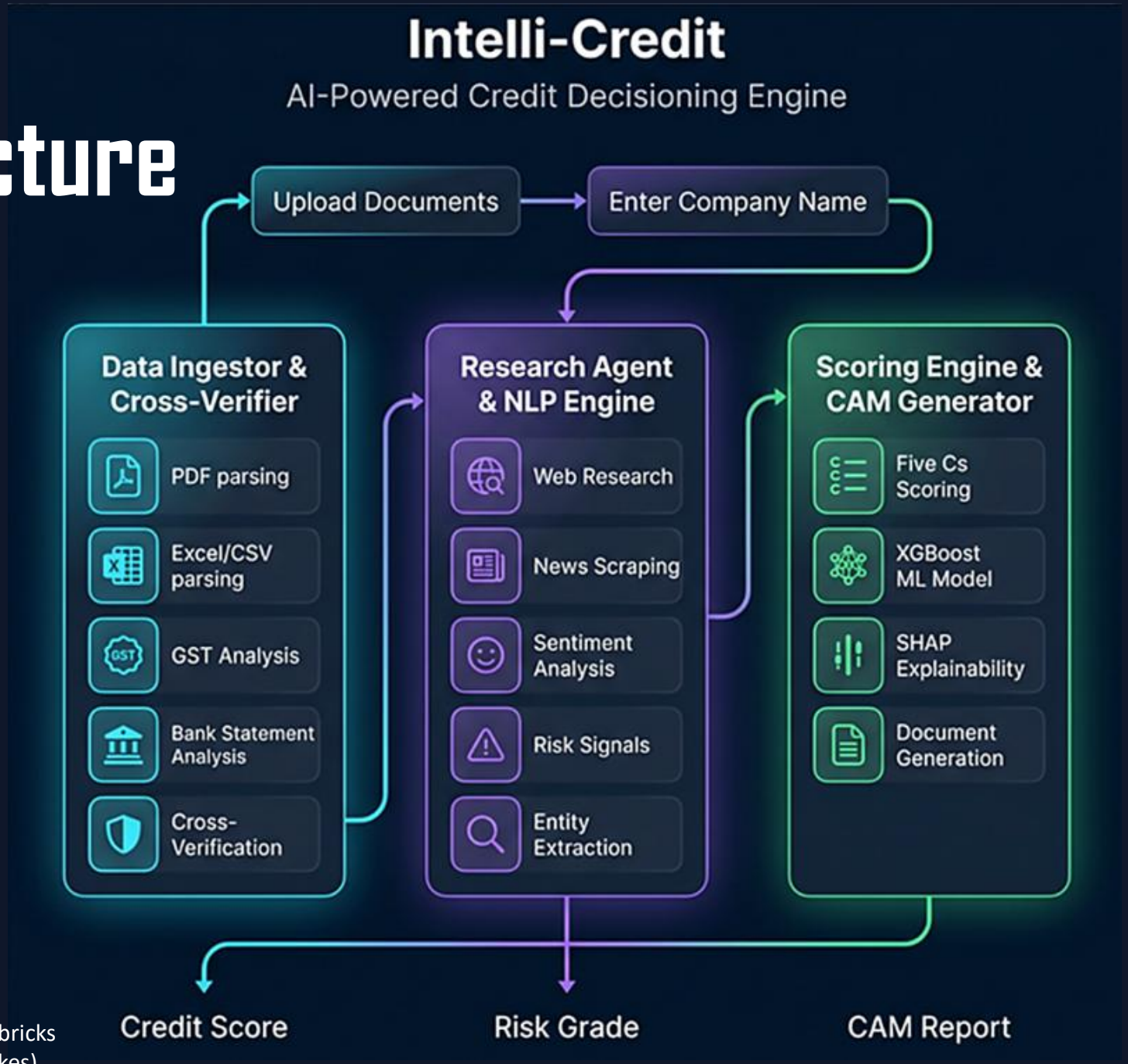
- Credit managers deal with a 'Data Paradox’ more data, yet weeks to process a single application.
- Stitching disparate structured data (GST, Bank Statements) and unstructured data (Annual Reports).
- Manual processes miss critical 'early warning signals' buried in unstructured text.
- Primary insights (site visits) and external intelligence (MCA, Legal) remain siloed.
- Result: Slow, manual and bias-prone credit appraisal.

# Our Solution: Intelli-Credit

- An AI-Native workflow automating corporate credit decisions end-to-end.
- Ingests multi-source unstructured & structured data seamlessly via Databricks-compatible pipelines.
- Performs deep 'web-scale' secondary research on promoters, sector headwinds and litigation.
- Synthesizes primary due diligence (Credit Officer notes) to instantly adjust risk scores.
- Generates a structured, committee-ready Credit Appraisal Memo (CAM) with Explainable AI.

# Technical Architecture

- Platform processes uploaded financial documents, verifies financial data, enriches it with external intelligence using NLP.
- Applies machine learning-based credit scoring.
- System outputs an explainable credit score, risk grade, and an automated CAM report to support faster and more reliable lending decisions.




Note: Built on Python/Flask for prototype speed; integrates with Databricks for production scale multi-source ingestion (e.g. GST/ITR/Bank data lakes)

# The 7-Step AI Pipeline

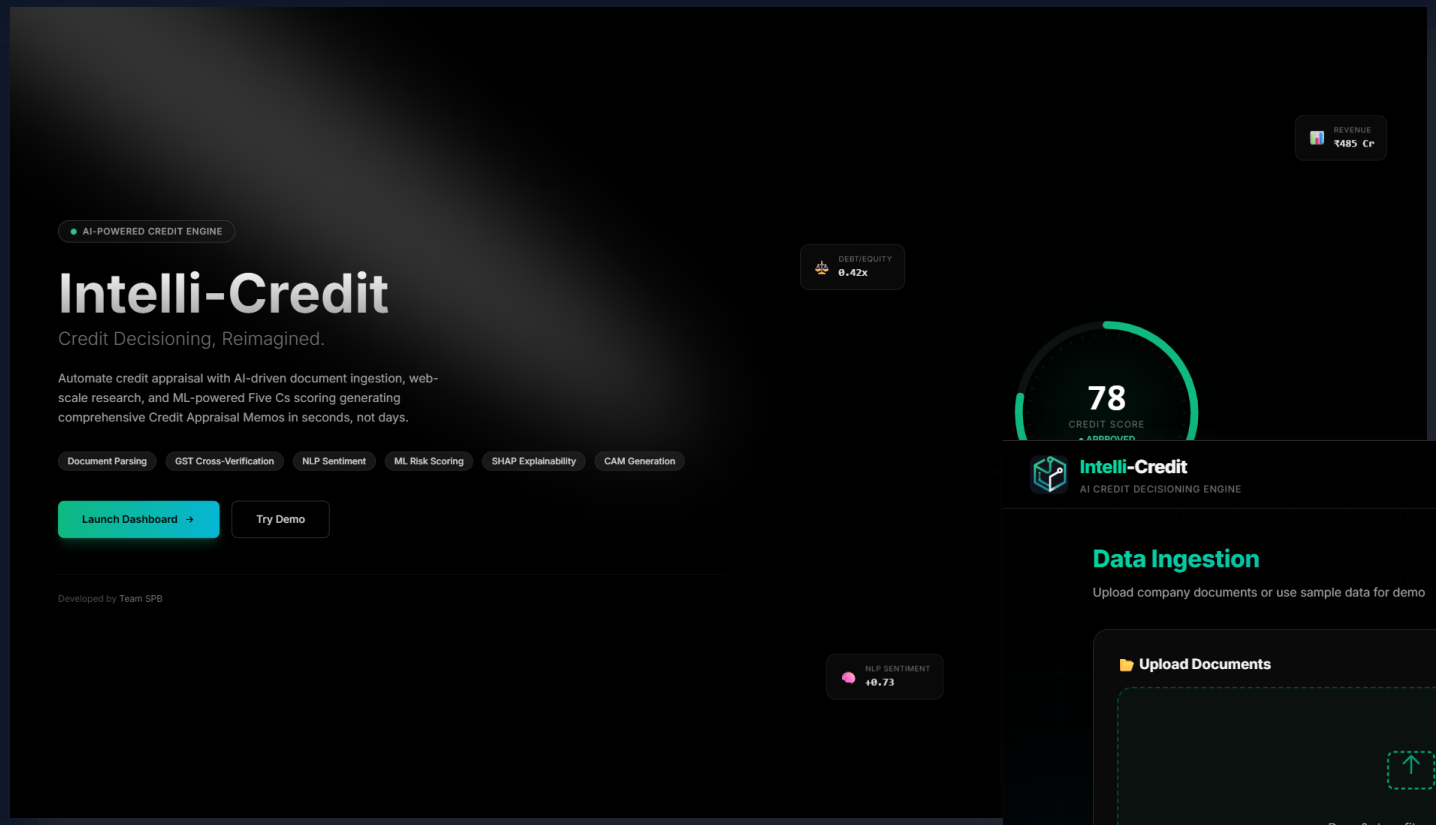
- Financial documents are ingested and cross-verified using GST and bank data.
- External research and NLP identify risk signals, after which machine learning models and the Five Cs framework evaluate creditworthiness.
- The system then generates a credit score, risk grade and an automated CAM report for decision support.

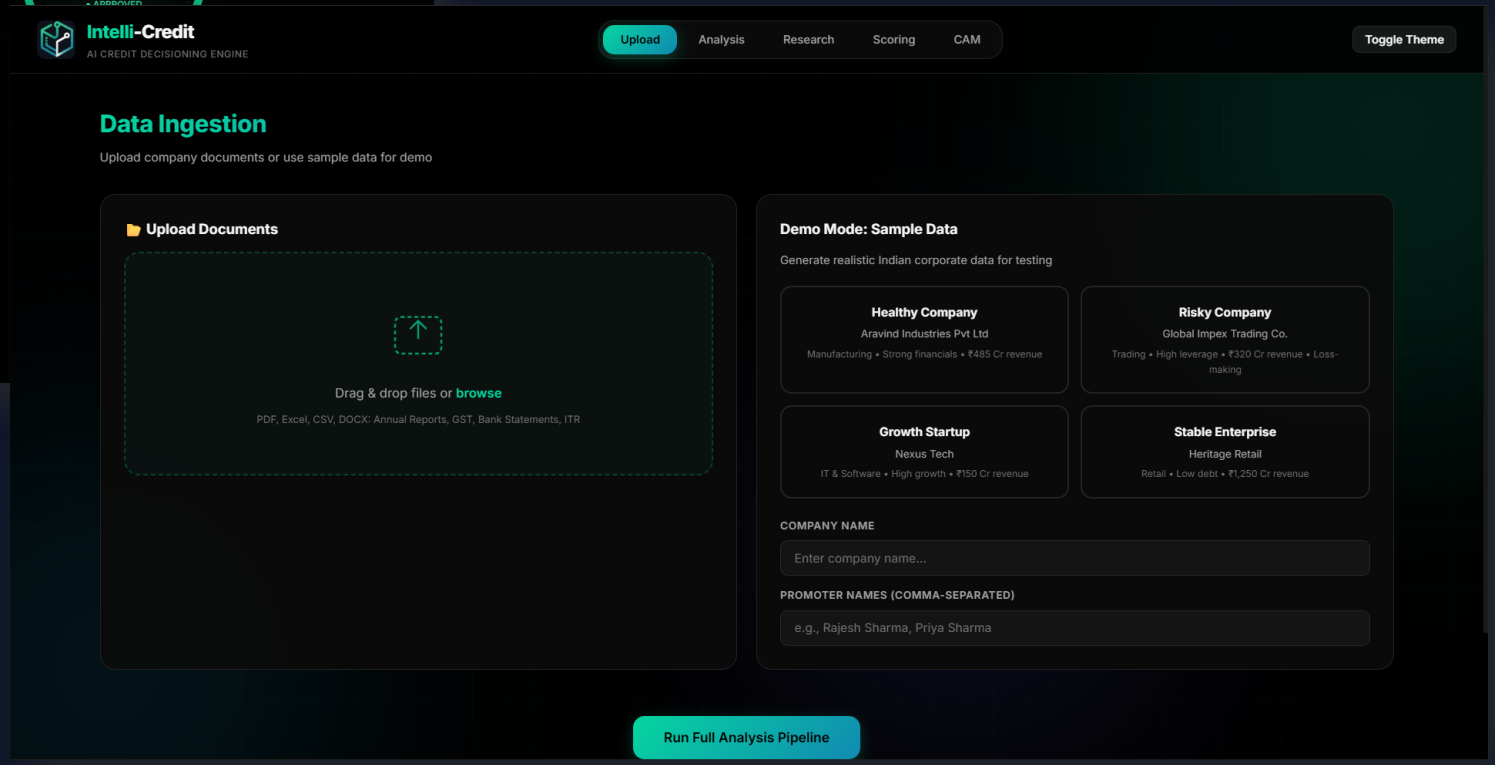


# The Three Pillars of Intelli-Credit

- 1. High-Latency Data Ingestor: Parses PDF reports and cross-leverages GST vs Bank Statements to detect 'circular trading'.
- 2. Research Agent ('Digital Credit Manager'): Web-scale crawling for news, MCA filings, and e-Courts litigation.
- 3. Recommendation Engine: Predicts limit/rate via ML and generates the Five Cs CAM.
- Primary Insight Integration: Qualitative factory/site visit notes dynamically adjust the AI model.

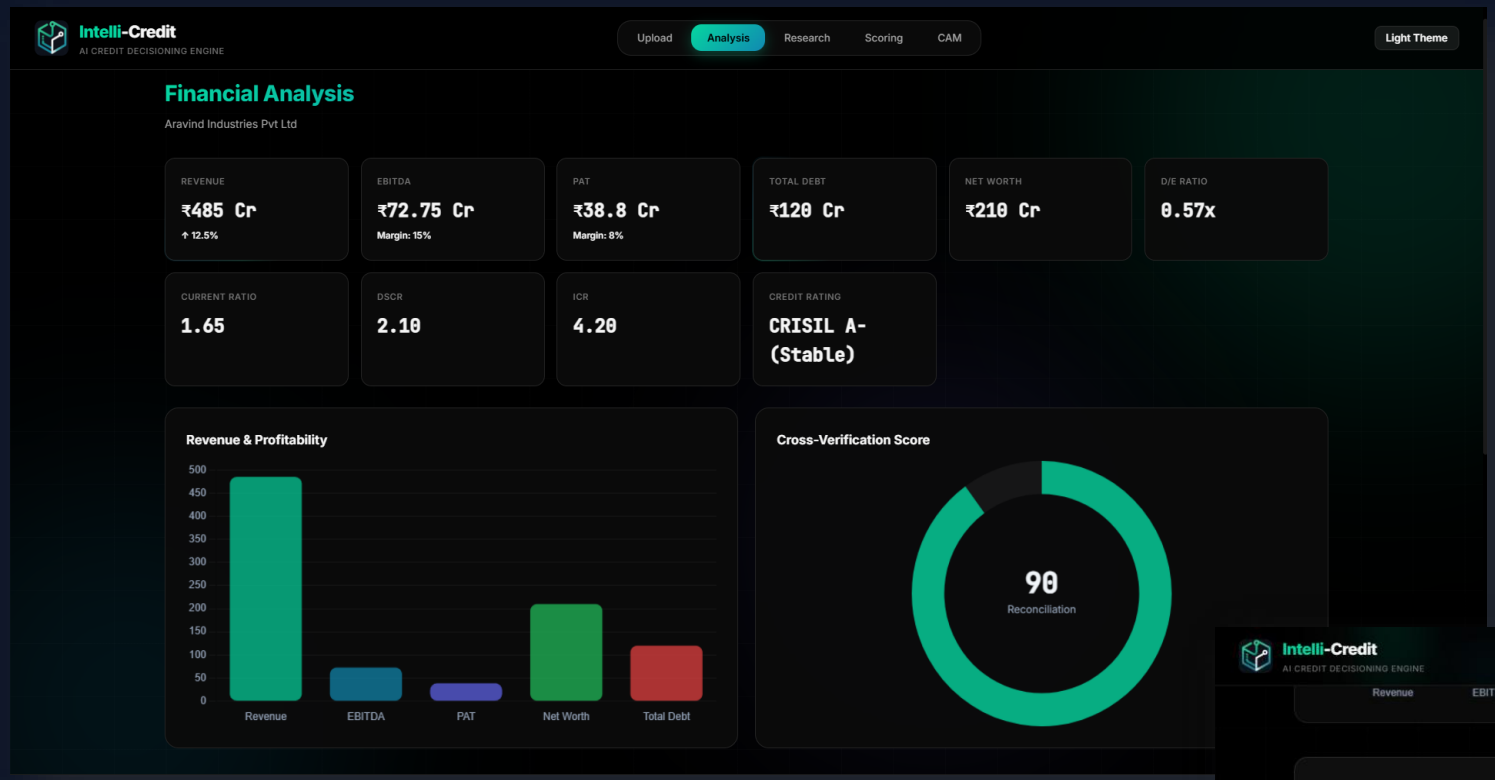
# User Interface Landing



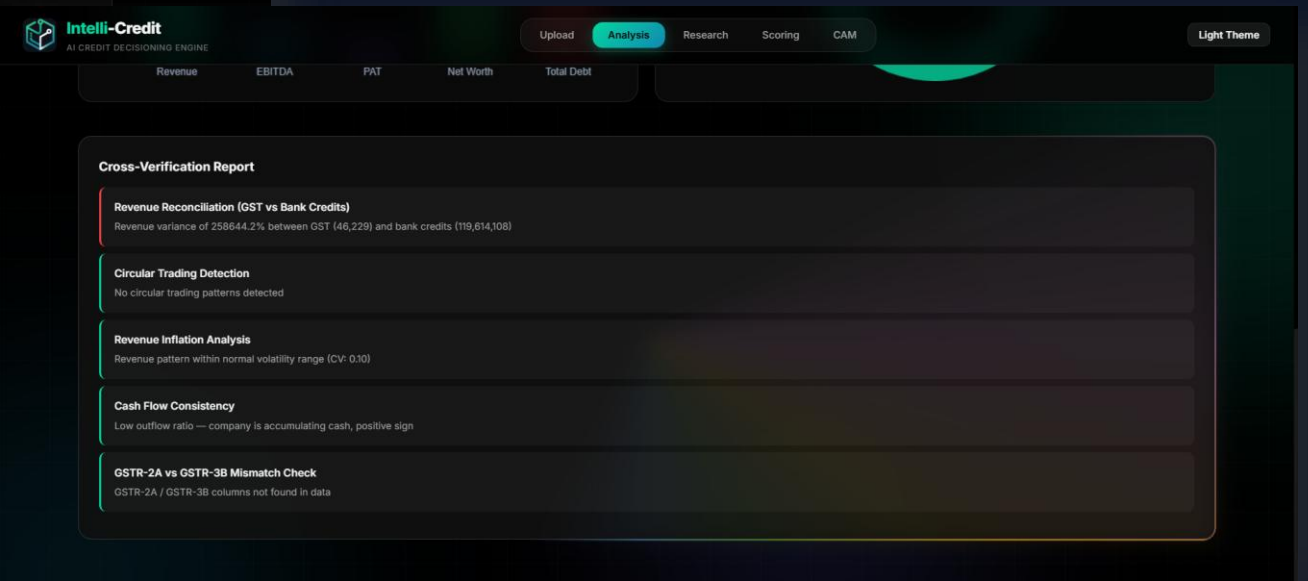




- Clean, modern interface for the Credit Manager.
- Accepts multi-format, unstructured and structured uploads.

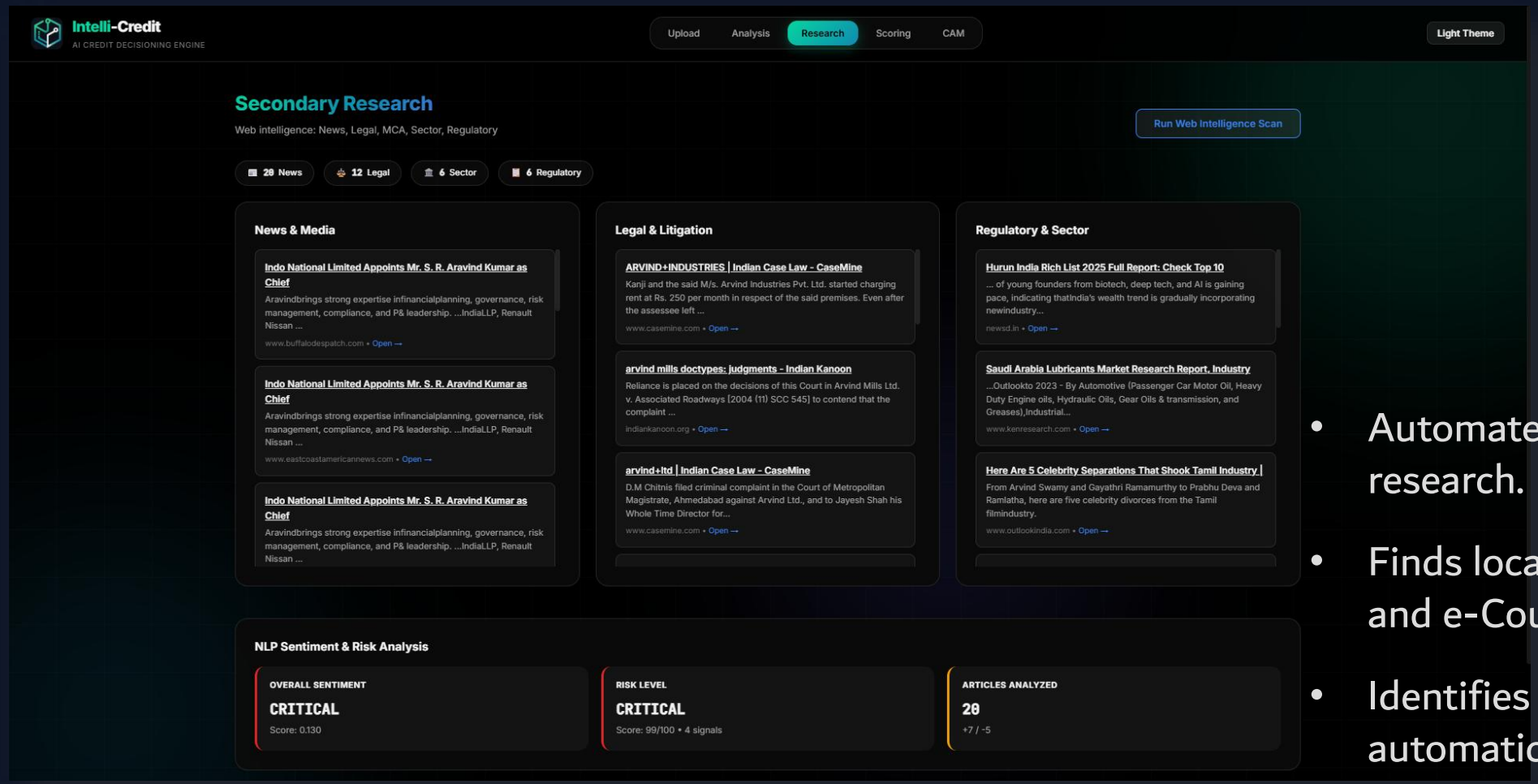
# Pillar 1: Data Ingestor & Synthesis




- Cross-leverages GST returns, ITRs, and Bank Statements to detect circular trading or revenue inflation.
- Specifically designed to identify 'circular trading' or revenue inflation.
- Handles India-specific nuances like GSTR-2A (purchase reconciliation) vs. GSTR-3B (summary returns) mismatches.
- Unstructured Parsing: Extracts financial commitments/risks from PDF annual reports, legal notices and sanction letters from other banks.

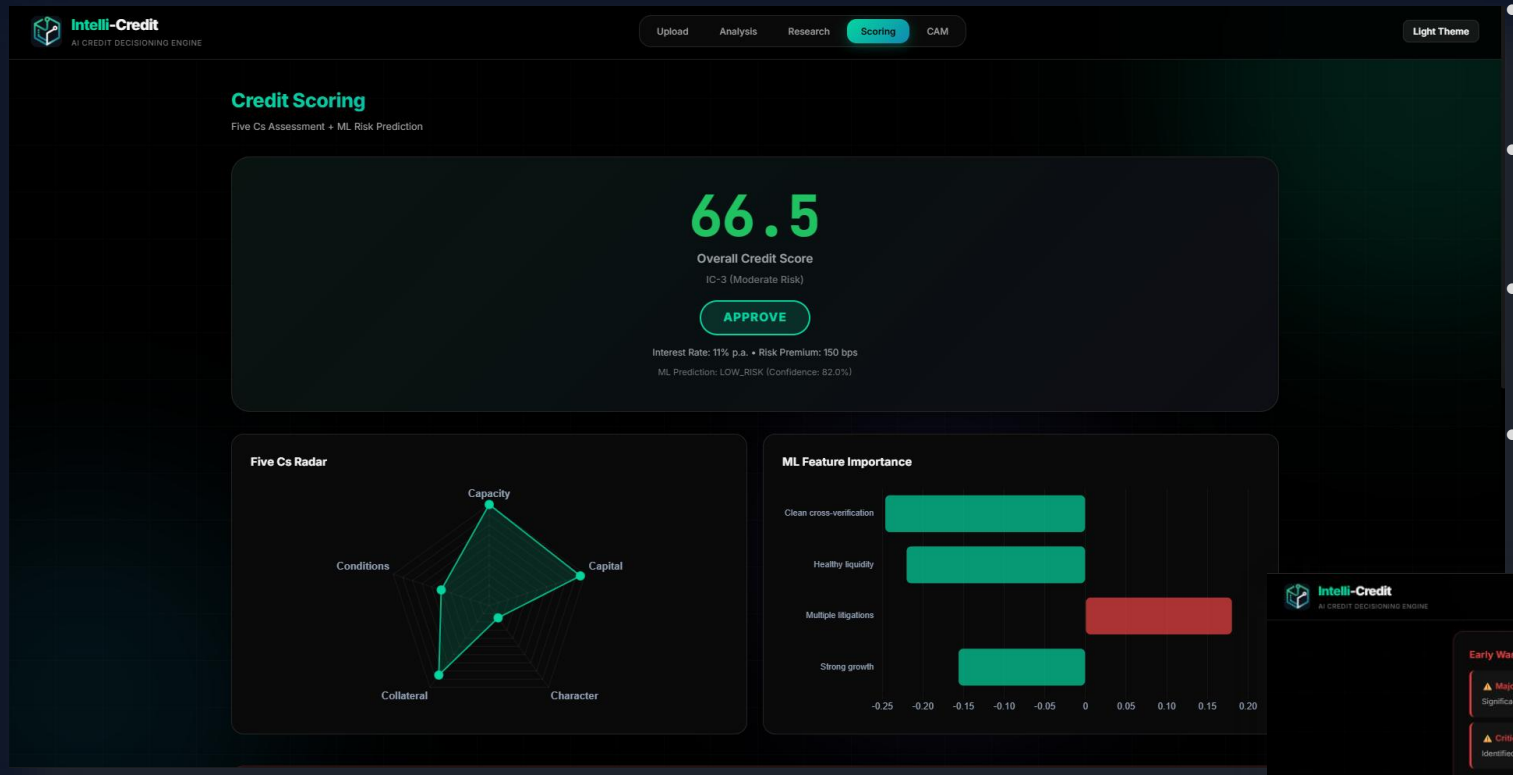


# Pillar 2: Research Agent

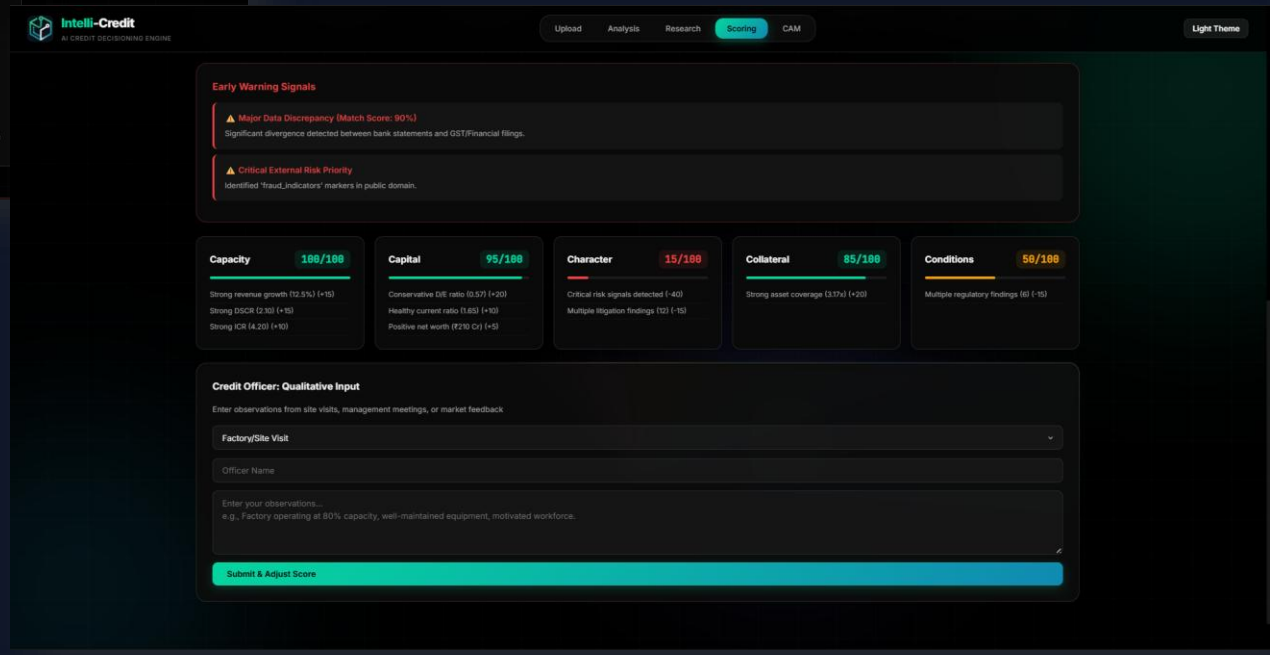



- Automated 'web-scale' secondary research.
- Finds local news, sector headwinds, and e-Court litigation history.
- Identifies 'early warning signals' automatically.

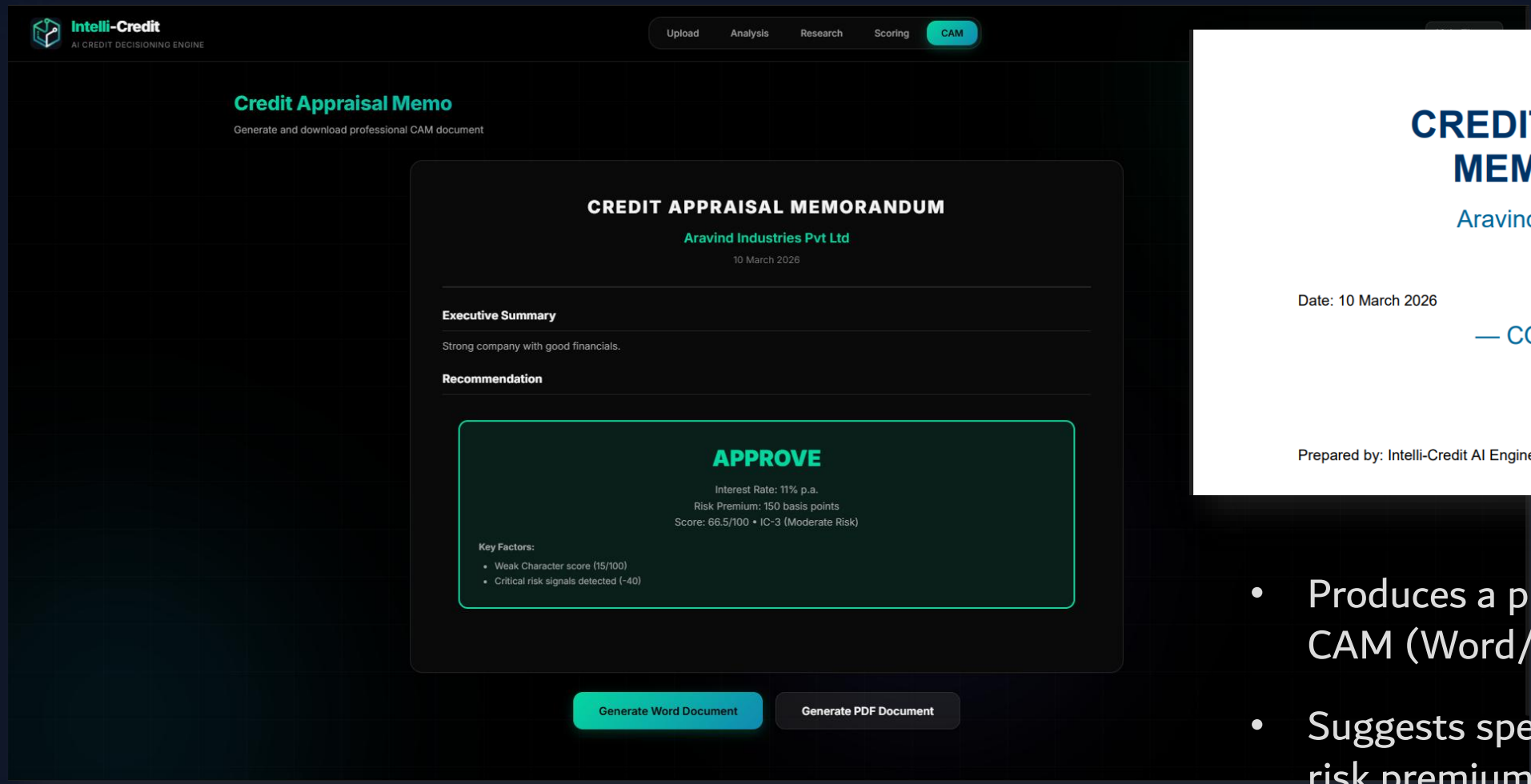
# Pillar 3: Recommendation Engine

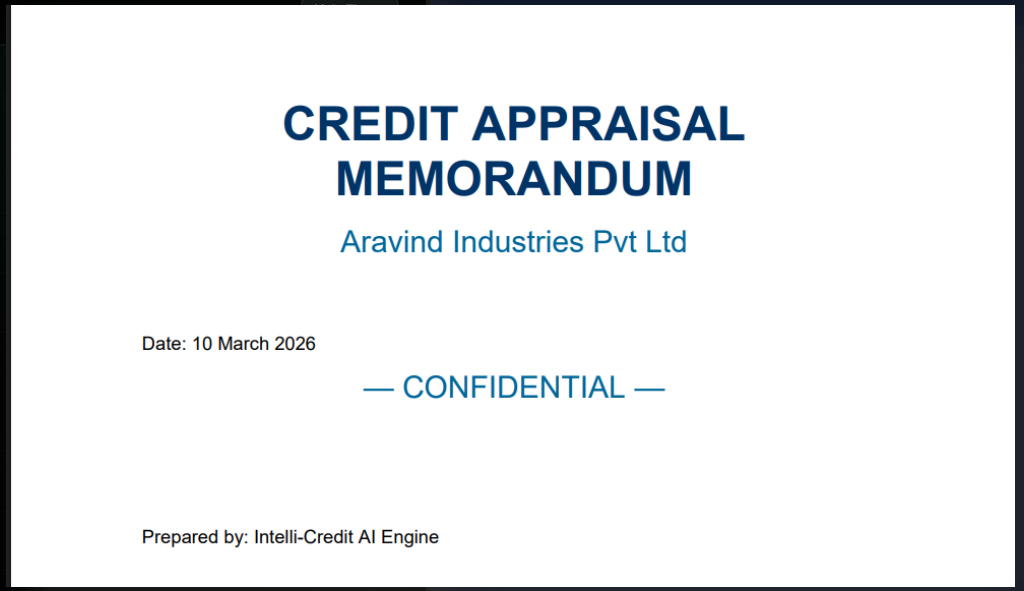



- Five Cs of Credit breakdown (Character, Capacity, Capital, Collateral, Conditions)
- Explainable AI: SHAP charts 'walk the judge through' the ML logic (not a black box).
- Primary Due Diligence: Input for Credit Officer site-visit observations and management interview notes.
- Suggests loan amount/risk premium with transparent logic, e.g. "Rejected due to high litigation risk found in secondary research despite strong GST flows."



# Final Output: Auto-CAM Generator




- Produces a professional, structured CAM (Word/PDF).
- Suggests specific loan amounts and risk premiums.
- Transparent explanations for Rejection or Approval limits.

# Technology Stack

- Frontend: HTML5, CSS3, Vanilla JS, Chart.js
- Backend: Python, Flask REST API
- Data Parsing: PyMuPDF, pdfplumber, pandas
- AI/ML Engine: XGBoost, SHAP, scikit-learn
- NLP / LLM: TextBlob, OpenAI GPT-4 API (for synthesis & sentiment)
- Document Generation: python-docx, reportlab

# Meeting the Core Evaluation Criteria

- Extraction Accuracy: High-fidelity parsing of Indian-context PDFs and CSVs, including scanned GST filings and messy unstructured docs (e.g., legal notices).
- Research Depth: Automated live-web search capturing local litigation & headwinds (e.g., MCA/e-Courts) that aren't in the provided files.
- Explainability: Complete transparency preventing the 'black box' syndrome via SHAP can the AI 'walk the judge through' its logic? (E.g., step-by-step rejection trace via litigation vs. GST).
- Indian Context Sensitivity: Handles GST vs. Bank reconciliations directly; understands India-specific nuances like GSTR-2A vs. 3B discrepancies and CIBIL Commercial reports for promoter scoring.

# Real-World Impact & Relevance

- **Innovation:** AI-native "Digital Credit Manager" detects early warnings (e.g., circular trading via GST-bank mismatches authorities uncovered Rs 7 lakh Cr in GST evasions FY21-FY25, including fake ITC claims). Businesstoday.com
- **Relevance to India:** Tailored for mid-sized corporates; reconciles GSTR-2A/3B for ITC fraud flags, integrates CIBIL for promoter risks cuts lending time from weeks to hours. LinkedIn , Telegraph.com
- **Impact:** Banks leveraging AI for early warnings reduce NPA slippage by 20-30%; potential ₹10Cr+ annual savings for 50-branch banks via bias-free decisions (e.g., Indian Bank's sub-2% NPA target aided by AI recovery).
- **Scalability & Broader Apps:** Databricks scales to 1000s of apps/day (Spark for data lakes); extends to MSME/UPI lending pilot-ready for Indian fintech’s.




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